

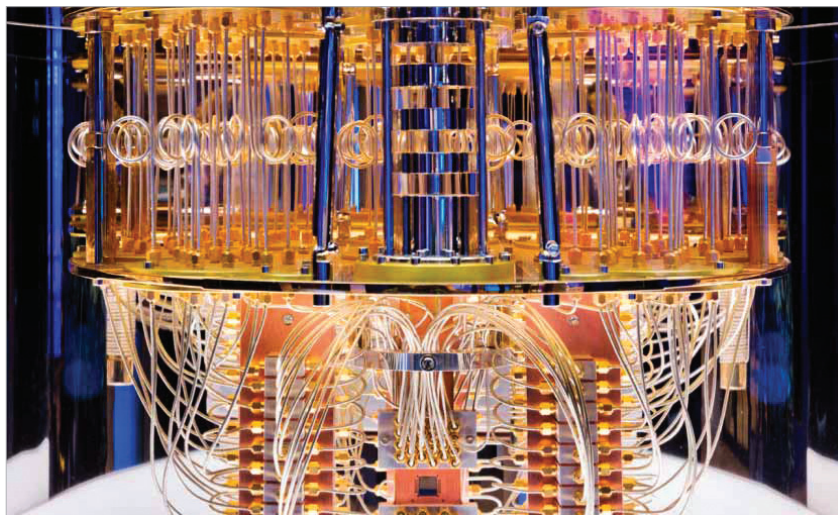


Figure 2: IBM's quantum computer

Quantum computing and performance factors

Quantum computing is one of the key areas of research for supercomputing and high performance applications. The traditional computing environment uses the concept of bits of 0 and 1, while quantum computing moves beyond bits and bytes and uses qubits.

In quantum computing, there is the possibility of using 0 and 1 simultaneously, which is not

possible in traditional computers. This superposition of qubits enhances the overall performance of the computing environment in which it is implemented. Such implementations are very useful for such real-time applications as satellite data analysis, avalanche predictions, geological data analytics, blockchain cryptography, and many others.

Google conducted an experiment in 2019 in which a 54-qubit quantum system was able to implement in a few minutes what would have taken a

classical computer 10,000 years using traditional bits and bytes.

Implementation of quantum computing on the cloud

A number of programming platforms are available on the cloud for using quantum computers remotely. These cloud platforms are provided by IBM, Google, Microsoft and many other technology giants.

IBM Q System One was the first commercial quantum computer to be based on circuit programming of quantum computing. This 20-qubit quantum computer was launched in 2019.

Quantum computing requires expensive hardware which is very complex to maintain. Currently, many cloud platforms provide open source quantum computers for researchers.

IBM Qiskit is one of the key open source programming platforms for researchers, scientists and practitioners of quantum computing, and is available at <https://github.com/Qiskit/qiskit>. Qiskit implementations can be done using Python programming. The command to install Qiskit using Python is:

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$ pip install qiskit
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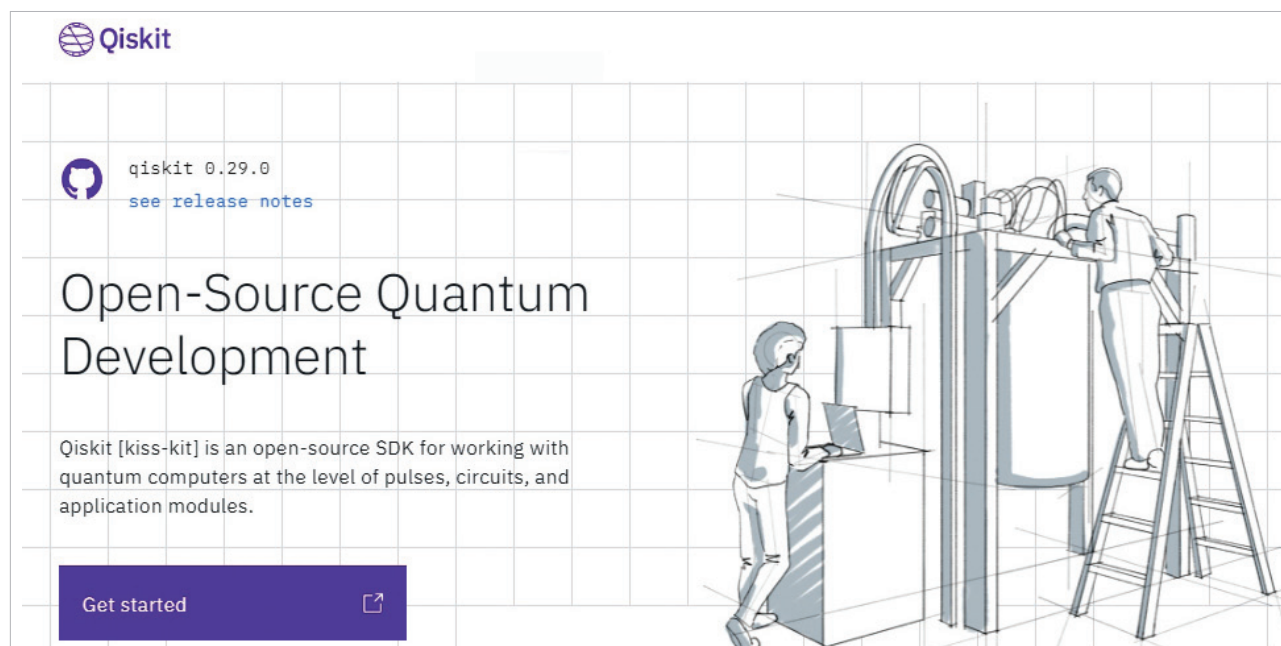


Figure 3: IBM Qiskit platform